

Progress Report: Image-Constrained Generative Modeling of Retinal Vasculature

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1. Current Progress

This project studies a rule-based generative model for retinal vasculature under image-derived structural constraints. At the current stage, the implemented pipeline uses fundus images only. The OCTA-based part proposed in the original project has not yet been incorporated, since those data have not yet been provided. The present results should therefore be understood as a fundus-constrained first-stage model, rather than the full multimodal framework described in the proposal.

The current codebase now supports an end-to-end workflow with three steps: (1) extraction of structural constraints from fundus images, (2) vascular tree generation under those constraints, and (3) quantitative comparison of multiple model variants. Compared with the previous update, the project has moved from an initial proof of concept to a working experimental pipeline.

2. What Has Been Implemented

The current implementation extracts the following information from each fundus image: retinal boundary, optic disc position and size, estimated macula position, vessel segmentation, vessel density map, and major vessel orientation near the optic disc. These extracted quantities are then used to define image-specific constraints for the branching generator.

Three model variants are currently compared:

Baseline: uses fixed manual parameters

Image-Constrained: uses parameters extracted from the fundus image

Density-Aware: extends the image-constrained model by incorporating the vessel density map into branching behavior

3. Main Results



Figure 1: Extracted structural constraints from representative fundus images, including optic disc location, macula estimate, and major vessel orientation.

Figure 1 shows that the extracted anatomical constraints are image-specific. Optic disc position, macula location, and the orientations of the superior and inferior arcades vary across images, so the generator is no longer driven by a single default geometry. This is the basis for the later constrained generation results.

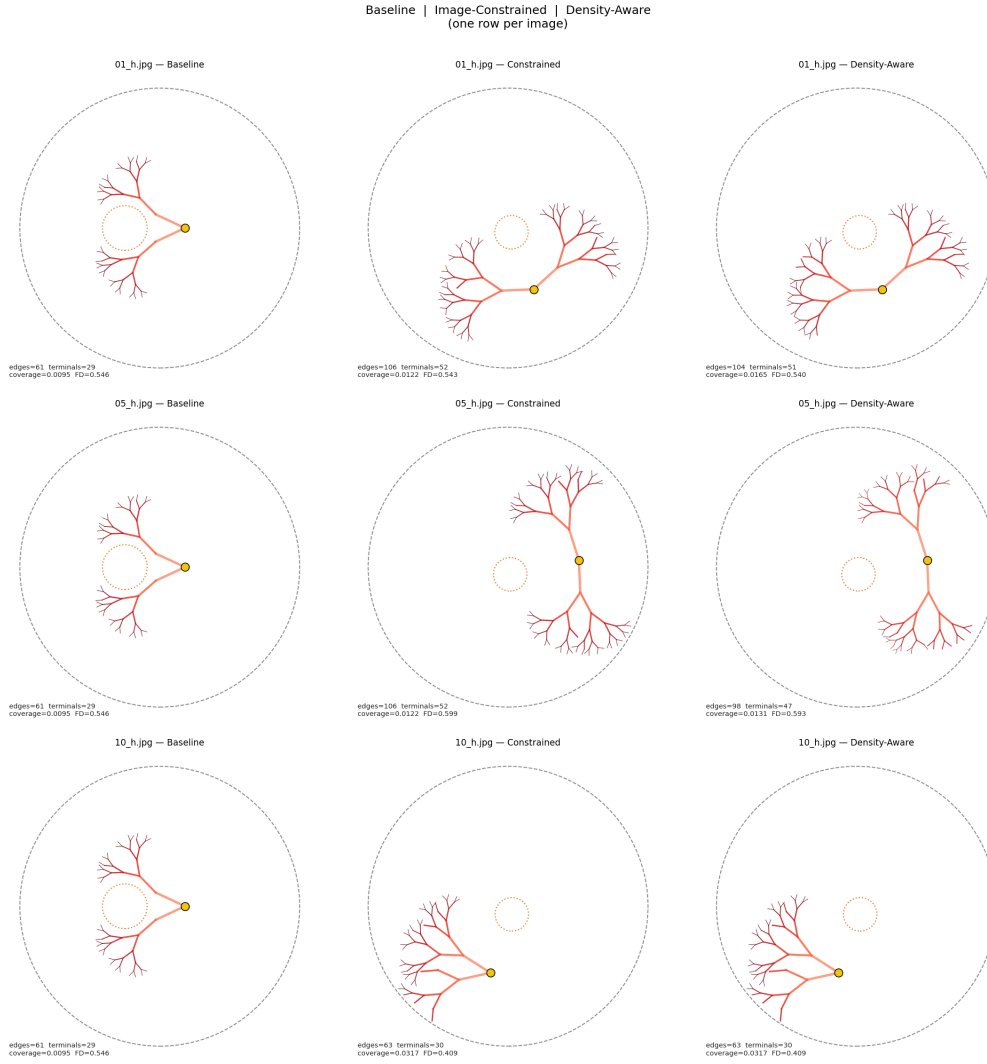


Figure 2: Comparison of baseline, image-constrained, and density-aware models across representative fundus images.

Figure 2 shows the main qualitative difference among the three model variants. The baseline model produces nearly the same generic vascular tree for different images, while the image-constrained model adjusts root location and major branching direction according to the extracted fundus anatomy. The density-aware model introduces additional local variation, but its effect is still moderate compared with the larger change produced by the image-constrained parameterization. Overall, the current model captures global anatomical layout more successfully than before, although local vessel spread is still limited.

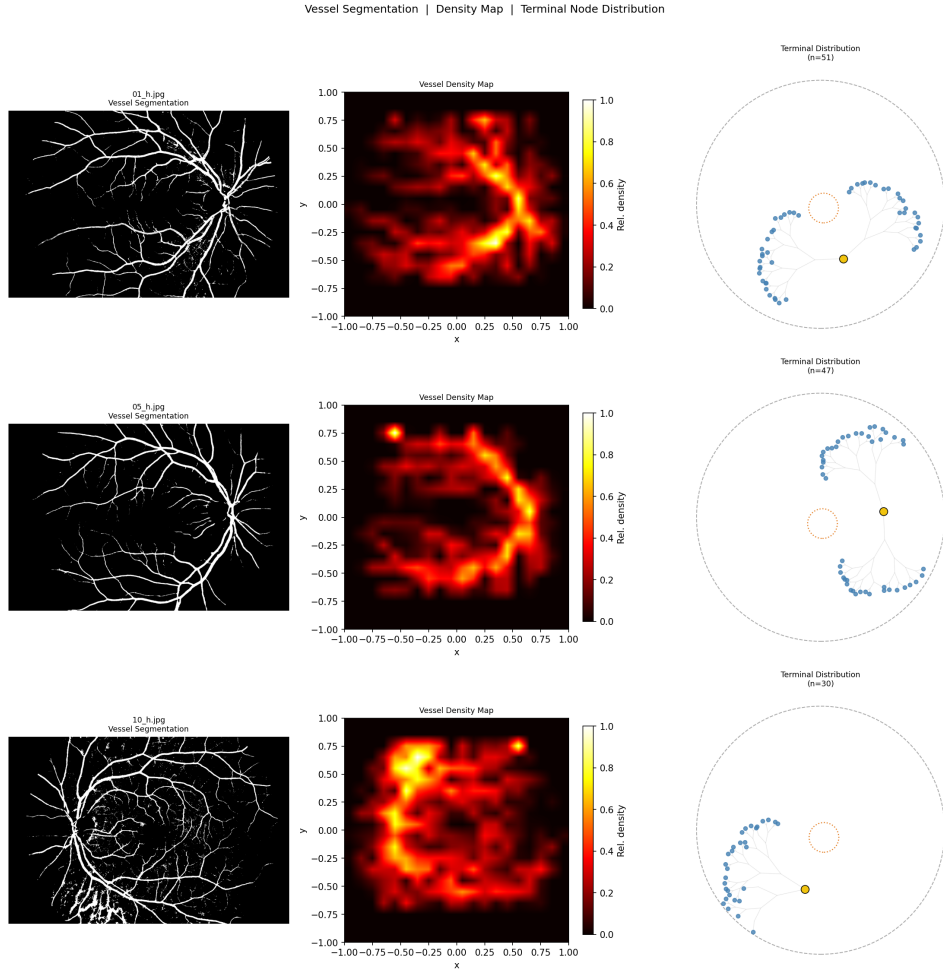


Figure 3: Comparison of vessel density maps and generated terminal-node distributions.

Figure 3 compares image-derived vessel density with the terminal-node distributions produced by the generator. The density maps capture large-scale asymmetry in the retinal vessel distribution, and the generated terminal nodes also show some directional bias. However, terminal nodes remain concentrated in relatively limited regions rather than spreading across the full area suggested by the density maps. This indicates that the current density-aware mechanism has begun to influence growth, but it is not yet strong enough to reproduce local vessel distribution realistically.

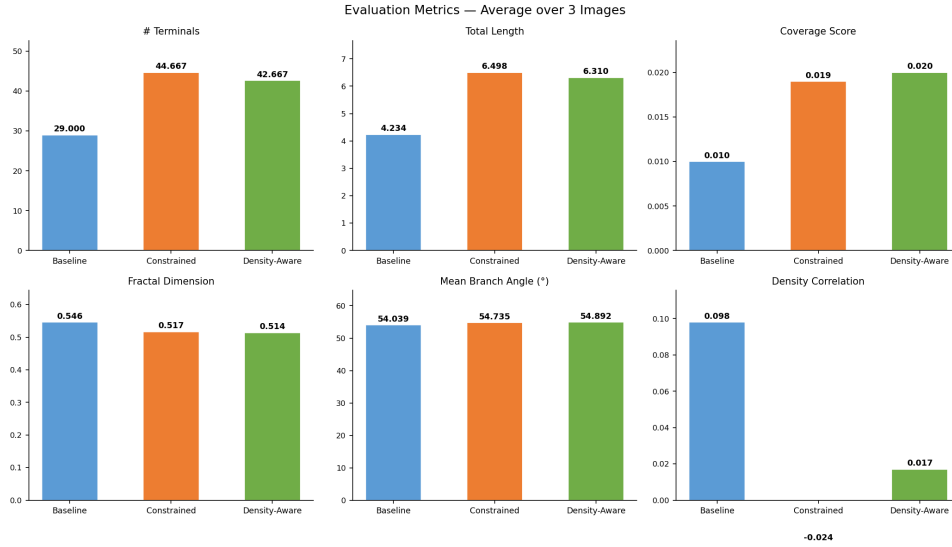


Figure 4: Quantitative summary of model performance averaged over three representative images.

Figure 4 confirms the same pattern quantitatively. Relative to the baseline, the constrained models produce more terminals, greater total vessel length, and better coverage. Averaged over three images, the number of terminals increases from 29.0 in the baseline to 44.7 in the image-constrained model, and coverage improves from 0.010 to about 0.019–0.020. The density-aware model gives the highest coverage score, but only a small improvement over the image-constrained model. At the same time, density correlation remains weak, indicating that the current density mechanism is not yet sufficient to match real vessel-density patterns closely.

4. Current Stage of the Project

At this point, the project has successfully established a fundus-based constrained generative framework. In other words, the model is no longer purely generic: it is now anatomically anchored to each input image through optic disc location, macula exclusion, and major vessel orientation.

However, the project is still at an intermediate stage. The current model is more successful at capturing global anatomical structure than local vessel distribution. The generated trees are visibly more image-specific than the baseline, but they do not yet reproduce the finer spatial spread of the real vasculature.

5. Next Steps

First, the density-aware branching rule needs to be strengthened so that local density affects terminal placement more directly, not only branching depth. This is the main modeling weakness exposed by the current results.

Second, the evaluation should be refined. In particular, the fractal-dimension calculation is still

quite rough, and a stronger topology- or distribution-based comparison would make the results more informative.

Third, once OCTA data become available, the next major extension will be to move from the current fundus-only constraint model toward the multimodal framework originally proposed. That would allow deeper vascular density and microvascular structure to be incorporated directly, rather than approximated only from fundus vessel segmentation.

6. Summary

In summary, the current stage of the project has completed the first fundus-based phase of the proposed framework. The code now supports image-specific constraint extraction, constrained vascular generation, and preliminary quantitative evaluation. The main result so far is that fundus-derived constraints substantially improve anatomical plausibility relative to a fixed baseline model. The main remaining challenge is to improve local vessel distribution and later extend the model to include OCTA-derived constraints when those data become available.